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The perceived criticality of different time headways is depending on velocity

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ABSTRACT

Time headway (THW) presumably represents a velocity-independent measure of safe following distances during car-following. However, studies using on-road data show that drivers decrease their THWs as velocity increases (e.g., Brackstone, Waterson, & McDonald, 2009). This contradicts recent findings of simulator-based studies suggesting independence of velocity and chosen THW (Siebert, Oehl, & Pfister, 2014; Siebert, Oehl, Bersch, & Pfister, 2017). This study aimed at investigating the relationship between velocity and subjective risk of different THWs using an experimental setup with a motion-based driving simulator. $N = 29$ participants followed a preceding vehicle with five different velocities ranging from 30 km/h to 140 km/h in urban, rural and highway environments. At each velocity level, drivers were instructed to follow with three different THWs (0.7 s, 1.1 s and 1.5 s). Subjective criticality ratings were analysed for the different combinations of these independent variables (IV1: velocity, IV2: THW). Drivers rated situations with short THWs as more critical when following with slower velocities compared to higher velocities. These results show that the criticality of theoretically velocity-independent THWs is indeed depending on velocity. These findings have both theoretical implications, such as a better understanding of drivers' car following behavior and risk acceptance, and practical implications as they may be used in the design of automated driving functions.

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1. Introduction

Car following is one of the most fundamental tasks during driving. Technologies such as Adaptive Cruise Control (ACC) allow drivers to hand off speed and acceleration to driver assistance systems that maintain a comfortable distance to preceding vehicles. One crucial parameter for these systems is an optimal time headway (THW). The THW is defined as the distance to the preceding vehicle divided by the speed of the following vehicle. It thus can be described as the time it takes until the following vehicle will pass the momentary position of the preceding vehicle. When deciding whether the following distance is safe or not, the main advantage of this measure over meter-based distances is its independence of velocity and that it allows for calculations of required decelerations that are not constrained to a specific velocity range (Winner, 2009). Driver trainings in different countries recommend thresholds for highly critical THWs (e.g., Germany: 1.8 s, US: 2.0 s; Michael, Leeming, & Dwyer, 2000; Underwood, 2005, p. 244). Still, drivers often choose THWs below this threshold, possibly rendering their behavior dangerous for themselves and other road users.

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It is thus of great interest to better understand the cognitive processes that cause drivers to adopt possibly dangerous following distances. Different factors that influence preferred THWss have been reported in the literature. This paper aims to highlight especially the role of velocity in the context of preferred THWs and subjective risk. We argue that velocity is a key factor for understanding appraisal process involved in the assessment of criticality of following distances, however, it is precisely in this regard the traffic observations, and experimental findings have provided mixed results. The remainder of the paper is as follows. Known influencing factors on chosen following distances are explained, considerations on the effect of velocity on preferred THWs are summarized, and the results of the study are reported and discussed.

1.1. Preferred time headways – influencing factors

Several influencing factors on preferred THWs have been reported. For example, studies show that drivers choose smaller THWs when they score high on the personality traits aggression (Maag, Benmimoun, & Neunzig, 2004) or sensation seeking (Heino, Van der Molen, & Wilde, 1992). Younger adults choose smaller THWs compared to older adults (Evans & Wasielewski, 1983; Ohta, 1993) and sleepy drivers tend to follow with larger THWs (van der Hulst, Meijman, & Rothengatter, 2001). On the other hand, situational factors also influence the preferred THWs. Drivers tend to increase their THW because of fog (Broughton, Switzer, & Scott, 2007; van der Horst & Hogema, 1993) or rain (Ichikawa, 2003). Interestingly, only older drivers tend to choose larger THWs at night compared to daytime (Kiefer et al., 1999). Traffic states also seem to influence preferred THWs as drivers tend to close gaps between vehicles during congested traffic, resulting in smaller THWs compared to free-flow traffic (Ayres, Li, Schleuning, & Young, 2001). Furthermore, drivers seem to be inconsistent in their day-to-day choosing of THWs (Brackstone, Waterson, & McDonald, 2009). In sum, several influencing factors on preferred THWs have been reported so far. The scope of this study, however, is on the influence of velocity on preferred THWs, as will be explained in the next section.

1.2. The effect of velocity and driving environment on time headways

Using instrumented vehicles, Brackstone et al. (2009) report that chosen THWs can be split by velocity roughly in two ranges. THWs stay approximately constant between 1.0 and 1.5 s above a velocity of 15 m/s (≈50 km/h), whereas THWs increase up to 2.2 s when the velocity is under this threshold. In line with this, several on-road studies investigating different velocity ranges report different preferred THWs. For high velocities, Knospe, Santen, Schadschneider, and Schreckenberg (2002) found a peak in the distribution of THWs at about 0.9 s on German highways. Brackstone, Sultan, and McDonald (2002) showed that 95.8% of all THWs of UK's highways fall under 2 s and 29.2% even under 0.8 s. Accordingly, Brackstone et al. (2009) report a maximum frequency of approximately 1 s on UK's highways. In contrast, in urban areas with lower velocities, larger THWs are found. Here, according to Piao and McDonald (2003), most common THWs range from 1 to 2.5 s with a peak at 1.9 s. Ichikawa (2003) consistently reports mean THWs in urban areas of 2.11 s. Comparing reported on-road THWs from studies conducted on highways versus those measured in urban areas suggests a clear effect of velocity on the drivers' preferred THWs. When driving slowly, drivers prefer larger THWs compared to high velocities even though the THW is commonly seen as a velocity-independent distance measurement. Still, one has to acknowledge a possible influence of the completely different driving environments (urban roads versus highway) on the driver behavior. To the knowledge of the authors, however, no experimental proof of a velocity effect on preferred THWs either in the same or different driving environment has been reported in the literature.

On the contrary, studies in driving simulators suggest a clear absence of a velocity effect on preferred THWs during car following scenarios. Based on a previous study from Lewis-Evans and Rothengatter (2009), Siebert, Oehl, and Pfister (2014) as well as Siebert, Oehl, Bersch, and Pfister (2017; study part one) report no significant effect of velocity on the subjective risk rating of different THWs that ranged between 0.5 s and 4.0 s. Contrary to the on-road studies described above, they did not investigate natural driving behavior and observable preferred THWs but the subjective risk of different following distance while longitudinal vehicle control was an automated system (by ACC). Still, this contradicts a possible mechanism that drivers choose smaller THWs when driving with lower velocities because they perceive the same THW as more critical compared with driving in higher velocities, as would be suggested by the above-cited on-road studies. Additionally, Siebert et al. (2017; study part two) asked participants in a free-driving simulator scenario to follow a preceding car with a risky or not comfortable distance and showed that the chosen THWs were independent of velocity. Older simulator studies similarly suggest that preferred THWs should be independent of velocity (Taieb-Maimon & Shinar, 2001; van Winsum & Heino, 1996). One possible explanation for a missing velocity effect in earlier studies could be the mostly generic driving environment that was independent of velocity in most studies. However, a realistic driving environment might, in fact, be essential for finding a velocity effect on criticality ratings during car following. In line with this, Siebert et al. (2014) presumed, but did not investigate, a possible influence of the driving environment on the subjective risk of each THW. For example, on rural roads with a recommended traveling speed of 100 km/h, the same THWs were rated as more critical for higher or lower velocities (50 km/h and 150 km/h) compared to the same THWs at the recommended velocity of 100 km/h. They argue that the typical traveling speed for a specific road type could be taken as being the safest and that deviations from the recommended speed increase subjective risk. In sum, a velocity effect on preferred THWs or subjective risk for different traveling velocities as suggested by on-road data has not been reported in controlled experimental settings so far.

1.3. The effect of velocity and brake distance on time headways

Besides on-road and simulator data, another basis for the argumentation of a hypothetical velocity effect on subjective risk of THWs can be established when collision probability (i.e., the situation's criticality) associated with different THWs and velocities are considered, because "the preferred time-headway of drivers in highway conditions is related to the likelihood of rear-end collisions" (Ayres et al., 2001, p. 827). Consider a car following situation with both cars holding approximately the same traveling velocity. Suddenly, the preceding vehicle decelerates, e.g., during an emergency brake maneuver. Rationally, a safe following distance would be regarded as when the following car still can avoid a collision via braking (Winner, 2009) taking into account brake reaction times as proposed by Lee, McGehee, Brown, and Reyes (2002) of 0.8–1.2 s. Taieb-Maimon and Shinar (2001) state: "In steady-state following – when the two vehicles are moving at the same speed – the time headway represents the time available to the following driver to reach the same level of deceleration as the lead vehicle in case it brakes. This time is independent of speed." (p. 160). Still, following physical calculations, it should, in fact, make a difference, especially for small THWs, if drivers follow with higher or lower speeds. To illustrate this, Fig. 1 depicts the development of the absolute distance in meters between a preceding and a following vehicle over time, starting when the preceding car decelerates. Beginning at t_0, the theoretical velocities, accelerations, and the distance between both vehicles are calculated in small steps (10 ms) by taking into account both vehicles' current velocities, accelerations, and current distance. Δ_{distance} between two timestamps $t-1$ and t (for example, how much closer the following car comes to the preceding car in meters within a timeframe of 10 ms) is dependent on the relative velocity between both cars at the first timestamp $t-1$.

$$distance_t = distance_{t-1} - \Delta distance_{(t-1,t),} \quad with \quad \Delta distance_{(t-1,t)} = \frac{v_{rel,t-1}}{\Delta_{t(t-1,t)}}$$

The absolute distance at the second timestamp is thus the distance at the first timestamp $t-1$ minus Δ_{distance}. Accordingly, the new relative velocity at the second timestamp is dependent on the relative deceleration at the first timestamp.

$$v_{rel,t} = v_{rel,t-1} - \Delta v_{rel,(t-1,t),} \ with \ \Delta v_{rel(t-1,t)} = \frac{a_{rel,t-1}}{\Delta_{t(t-1,t)}}$$

The graphs are based on a scenario with an initial THW of 0.7 s, a moderate to strong maximal deceleration of both vehicles of 6 m/s^2 following Winner (2009), a deceleration phase of the preceding car of 2 s (including a phase where maximum deceleration is building up, thus deceleration is not constant for the full two seconds) and a reaction time of the unprepared following driver of 1.2 s until the brake reaction triggers deceleration following Lee et al. (2002).

In this scenario, the same THW leads to collisions (that is a distance smaller zero) for velocities of 30 km/h and 50 km/h, but not for higher velocities. Notice, that all other factors are held constant. In this context, an important measure for analyzing the risk of rear-front accidents is the so-called time-to-collision (following Winner, 2009, p. 527):

$$t_{tc} = \frac{distance}{v_{rel,}}$$

Here, only the following distance in meters predicts the objective criticality of a driving situation, but not the THW. This shows that – in a strictly physical sense – the same THW (which is an already velocity-dependent distance measure) results in a higher analytical probability for a collision when velocity is low compared to when velocity is high. The main reason for this is that the same THWs result in different distances in meters for different velocities which leads to different probabilities of collisions. This analysis assumes a fixed value for deceleration regardless of the driving velocity. However, the deceleration has been shown to be dependent on velocity and the brake force, and intensity is often adjusted to different velocity ranges (van Winsum & Heino, 1996). Thus, the prediction of this analysis will differ to a certain extent from realistic driving.

From a psychological point of view, both the Risk Homeostasis Theory (Taylor, 1964; Wilde, 1982) and the Risk Allostasis Theory by Fuller (2008) would also deduce a velocity effect on preferred THWs and subjective risk. The Risk Homeostasis Theory states that drivers seek an optimal anxiety level based on the objective risk of being involved in an accident. If the above-reported different accident probabilities for different velocities – under constant THWs – are available to the driver (Wilde, 1982), a change of behavior concerning an adaptation of the THW or higher subjective risk ratings should be found. Risk Allostasis Theory (Fuller, 2008) predicts that the driver will adapt his behavior (here, the preferred THW) to the subjective risk, i.e., the risk of the situation given the drivers own capabilities to cope with its demands. Following this theory, a higher subjective risk for lower driving velocities would explain larger THWs for lower driving velocities. Importantly, driving demands are higher in city traffic (e.g., crossings, pedestrians on the street, bicyclists, traffic lights) than on the highway while the driver's capabilities remain constant at least in the short term. Since different velocity ranges are usually associated with differing driving environments, an effect of velocity on subjective risk ratings of THWs should also consider changes in the environment, additionally to the objective accident risk as presented in Fig. 1. Taken together, the combination of physical calculations with theories from traffic psychology would predict a velocity effect on preferred THWs and subjective risk.

One open question is still whether the subjective risk increases continuously with smaller THWs (Fuller, 2005) or whether subjective risk remains almost zero until a certain threshold-THW from where it increases abruptly. This threshold has been proposed to be around a THW of 2.0 s (Lewis-Evans, De Waard, & Brookhuis, 2010; Siebert et al., 2014). As soon as drivers

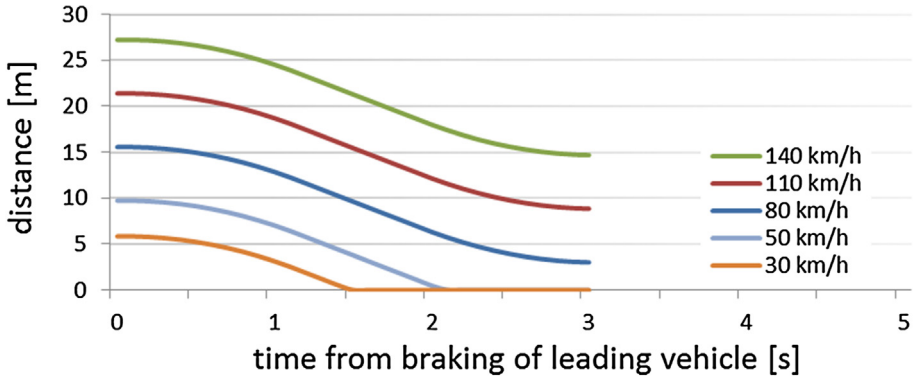

Fig. 1. Distance in meters between the preceding and the following car starting from braking of the preceding car for the different velocities and one standardized THW of 0.7 s at the start. $\text{Deceleration}_{max} = -6.0\ m/s^2$ (both vehicles), $\text{deceleration}_{preceding_vehicle}$ for 2.0 s, reaction $\text{time}_{following_vehicle} = 1.2$ s.

follow with shorter THW criticality increases but not before. The underlying problem here is whether the perceived risk of being involved in an accident will increase gradually or whether there will be a threshold relationship between perceived accident risk and objective influencing factors. Since besides THW also velocity contributes to the objective probability of being involved in an accident it is therefore as well an open question if the here suggested velocity effect on subjective risk is linear or follows a step function.

1.4. Research question

Even though on-road studies indicate that drivers choose different THWs in different velocity ranges, experimentally controlled studies do not show any velocity effect on the preferred THW or on subjective risk ratings. Further support for a velocity effect can be found in the combination of physical considerations and classical theories from traffic psychology. Hence, the aim of the present study was to investigate whether the velocity effect on subjective risk ratings during car following scenarios can be shown in an experimental setting. Of particular interest was the drivers' subjective estimation of the situation's criticality since it would contribute to the deeper understanding of the psychological mechanisms that could explain traffic observations (i.e., if a velocity effect on subjective risk can be found, this could explain the velocity effect on preferred THWs found in on-road data). Several aspects could have caused the missing velocity effect in previous driving simulator studies and were therefore considered thoroughly in this study: First, a motion-based driving simulator was chosen to allow for distance estimations that have been shown to be comparable to estimations while driving on a test track (Purucker, Rüger, Schneider, Neukum, & Färber, 2014). Second, because driving environments differ between velocity ranges in real driving (urban, rural or highway), the driving environment was adapted to the different velocity ranges in the study, a detail rarely considered in the methodology of previous driving simulator studies (e.g., Siebert et al., 2014, used the same driving environment with no roadside objects except for sparsely placed trees). Third, the driving task was chosen to be more realistic compared to other studies in a twofold manner. On the one hand, participants in the study of Fuller, McHugh, and Pender (2008) rated videos and participants in the study of Siebert et al. (2014) used in the main part of the experiment a ACC-system. Thus, the subjective risk ratings were based on a fixed THW to the preceding car, even though participants were still actively involved in the driving car via the lateral steering. In this study, we thereby present an approach to keep the driven responsible for the driving task while still being able to manipulate independent variables for the experiment (here, the THW). On the other hand, car following scenarios of other studies were not completely realistic since the preceding vehicle never braked (Fuller et al., 2008; Lewis-Evans & Rothengatter, 2009; Siebert et al., 2014). In an on-road car following situation this is often the case. Preceding cars accelerate or decelerate and sometimes even strong deceleration rates as, for example, during an emergency brake can occur. If the preceding car in a simulator study never brakes the influence of velocity on the probability of a collision as presented in Section 1.3 might be rendered irrelevant by the drivers thus hiding the velocity effect on subjective risk.

In sum, these methodological considerations for the study were expected to allow the observation of two effects: (1) smaller THWs should be rated higher regarding subjective risk of the car following scenarios replication of Siebert et al. (2014) and (2) the velocity effect should be found both as a general trend over all velocities as well as within the same driving environment.

2. Material and methods

2.1. Participants

$N = 29$ participants were recruited from a panel in a German town. There were no prerequisites or exclusion criteria other than normal or corrected vision and a valid driving license. Gender and age covered different samples of the German

population as well as different driving experiences regarding urban, rural and highway roads. 13 participants were female. The mean age of all participants was 38.12 years (*SD* = 12.99, minimum 22 and maximum 61). Participants received 25–30 € (approximately 30 $) for taking part in the study, depending on the individual length of the experiment.

2.2. Materials

The study was conducted using a motion-based driving simulator to increase the validity of the drivers' velocity estimations (Kemeny & Panerai, 2003; Purucker et al., 2014). The integrated vehicle's console contained all the necessary instrumentation and was identical to a production type BMW 520i with automatic transmission. To simulate a realistic steering torque, a servo motor based on a steering model was used. The motion system used six degrees of freedom and could briefly display a linear acceleration up to 5 m/s^2 or 100°/s2 on a rotary scale. Three LCD projectors were installed in the dome of the simulator and provided the projection that was a 180° screen image. Exterior and interior mirrors were implemented as LCDs. In addition to information about the driver behavior (input), also the simulated position, velocity, and acceleration of the test vehicle, as well as data concerning the adjacent traffic were saved to a data set with 100 Hz.

Table 1 summarized the experimental design with two independent variables (velocity and THW). The car following scenarios for five different velocity ranges (30, 50, 80, 110 and 140 km/h) were designed using the driving simulation SILAB® of the Wuerzburg Institute for Traffic Sciences (WIVW). Driving environments were adapted to the velocity ranges, resulting in the environments "urban" (30 and 50 km/h), "rural" (80 and 110 km/h) and "highway" (140 km/h). Only one lane per direction (separated by dashed lines) was implemented, except for the highway scenario (140 km/h: two lanes separated by dashed lines per direction, one solid separation between both directions). After all scenarios, one side street allowed the preceding vehicle to leave the main road into a small side street or a highway exit, while participants had to stay on the main road. Otherwise, the scenarios did not include any side streets during car following. Also, no additional objects like pedestrian crossings or traffic lights were positioned along the main road. In urban areas, buildings and pedestrians were added to the scenery, whereas in rural and highway roads hills, grass or field areas were used as the environment. Opposite lanes were filled with random traffic. The own lane contained only the own and one preceding car.

For each velocity range, participants were instructed to follow a preceding vehicle with three different THWs (0.7 s, 1.1 s, and 1.5 s). The instruction of the respective THW was realized via a semi-translucent enhanced reality strip (ERS) superimposed on the road surface (see Fig. 2). The end of the ERS indicated the optimal following distance to be maintained, and participants had to keep the end of the ERS under the preceding vehicle. As long as the resulting THW was within a tolerance range of ±0.1 s, the color of the ERS was rendered grey. A yellow ERS indicated an excessive following distance while a blue strip signalled a following distance that was too close. To maintain the natural driving experience and decrease the task demand of keeping an ideal THW, participants were instructed that the ERS indicated only the optimal THW and that slight deviations (resulting in color changes to yellow or blue) were expected during the experiment.

At the beginning of each scenario, participants had to slow down slightly until the preceding vehicle had accelerated up to the respective target velocity of the scenario. Each situation lasted approximately 60–90 s. After each situation, the preceding vehicle turned right into a side street or a highway exit, while participants were instructed to keep straight and follow the main road. The five velocity blocks, as well as the three THWs per velocity block, were completed in randomised order. Between the scenarios, other traffic scenarios were included (e.g., turning on conjunctions, driving on empty roads). In additional scenarios (all driving environments), the preceding vehicle braked in car following scenarios similar to the above-described ones to create a more natural driving task with not too static preceding vehicles. The probability of the preceding vehicle braking was constant over all conditions and, importantly, participants could not anticipate brake events by any visual cues (e.g., traffic lights).

After each car following scenario, participants were asked to evaluate the following distance regarding the subjective risk as proposed by Fuller (2005) described in Section 1.3. Thus, drivers explicitly had to evaluate the situation and report their subjective estimation of the risk level based on the probability of losing control over the vehicle, for example, due to a collision. In line with Siebert et al. (2014), driver judgments were based on the scale for evaluating the criticality of drive and traffic situations (Neukum, Lübbeke, Krüger, Mayser, & Steinle, 2008). The scale represents a modified version of a rating scale that was developed to categorize lateral dynamics disturbance, e.g., wrong steering angle or overlapping steering torques (Neukum & Krüger, 2003). The rating process was twofold: First, drivers had to choose one of five verbal categories before they could specify a tendency to each category adjoining to the chosen one. The categories and the adjacent numbers were: "nothing noticed" (0), "harmless" (1–3), "unpleasant" (4–6), "dangerous" (7–9), and "uncontrollable" (10). Therefore, participants could rate an "unpleasant" distance with a tendency to "harmless" as "unpleasant (4)". The significant advantage of this category scale lies in the unambiguous definition of a subjective tolerance limit. This limit is clearly defined by

Table 1
Independent variables for the 2 × 2 within-subjects design.

Independent variable	Conditions
Velocity	30 km/h, 50 km/h, 80 km/h, 110 km/h, 140 km/h
THW	0.7 s, 1.1 s, 1.5 s

Fig. 2. Instruction of the different time headways via the colored strip.

describing the categories and the instructions. It specifies unequivocally in which category the situation must be placed that is experienced as safety-relevant by the driver (i.e., judgment "dangerous" with more than 6 points on the scale). Thus, the scale provides a precise distinction between critical and only "unpleasant" situations. Participants were informed that they could use the scale to rate the criticality of situations during the experiment. They were explicitly instructed to consider the verbal anchors and only afterward adjust their final rating to a specific number. For each rating, the instruction was standardized to the question "Please rate the situation on the scale". This scale focuses directly on perceived criticality of driving situations. Importantly, it directly links the criticality to subjective risk experience via its verbal anchors. Following Siebert et al. (2014), the scale was regarded as a valid instrument for measuring subjective risk experiences and situation criticality.

2.3. Procedure

All participants were already registered in the panel-database including demographic data as well as driving experience. After arriving at the simulator, a short introduction to the testing procedure was given in an extra room. The drivers were informed about the driving task and the duration of the study and gave informed consent. Then, participants were seated in the driving simulator and familiarized themselves for 10 min with the driving simulator on a test track while following a preceding vehicle using the ERS (see Section 2.2). Drivers were naïve regarding the study purpose. As a cover story, they were told that the ERS would outline a new driving assistance system for car following. The five velocities were blocked and completed in randomized order. Within each velocity block, the three different THW conditions were also completed in randomized order. After each car following situation, participants rated the subjective criticality of the situation. Between car following episodes, other traffic scenarios were included in the simulator run, like passing of intersections and the already mentioned brake situations, where the preceding car decelerated mildly or moderately forcing the driver also to adapt his velocity and reduce the traveling speed. Participants were allowed to take breaks between the velocity blocks.

3. Results

In total, 1.8% of all scenarios (8 out of 435) had to be excluded from the descriptive and statistical analysis, because drivers followed with considerably smaller or larger THW than instructed (cutoff: mean deviation from ERS-instructed THW greater than 0.2 s). Missing values were replaced by the median of the combination of factors. The resulting mean ratings for the different velocities and THWs are plotted in Fig. 3. Note that the ratings could range between 0 and 10 and participants were instructed that the threshold between "uncomfortable" and "dangerous" THW was to be placed between 6 and 7 points. The threshold between "harmless" and "uncomfortable" was located between 3 and 4 points.

Over all velocities, drivers rate THWs of 0.7 s as most critical ($M = 6.12$, $SD = 1.85$), 1.5 s as least critical ($M = 2.83$, $SD = 1.53$) and 1.1 s as between both ($M = 4.27$, $SD = 1.83$). Even in situations with THWs of 0.7 s, mean criticality ratings exceed the threshold to "dangerous" situations only for velocities of 30 km/h. In all other velocity conditions, criticality ratings with THWs of 0.7 s are not exceeding an "uncomfortable" level. THWs larger than 0.7 s are mostly rated as only "uncomfortable" (1.1 s) or "uncomfortable" to "harmless" (1.5 s).

As expected, mean ratings in each THW category vary between different velocities (see Table 2). Mean ratings range from 7.41 (30 km/h) to 5.37 (140 km/h) for THWs of 0.7 s, from 5.00 to 3.93 for THWs of 1.1 s and from 3.62 to 2.20 for THWs of 1.5 s. The same THW is rated less critical for higher and more critical for lower velocities. However, this relation between velocity and criticality rating is not entirely linear. Participants rate velocities of 50 km/h as slightly less critical than velocities of 80 km/h for all THWs.

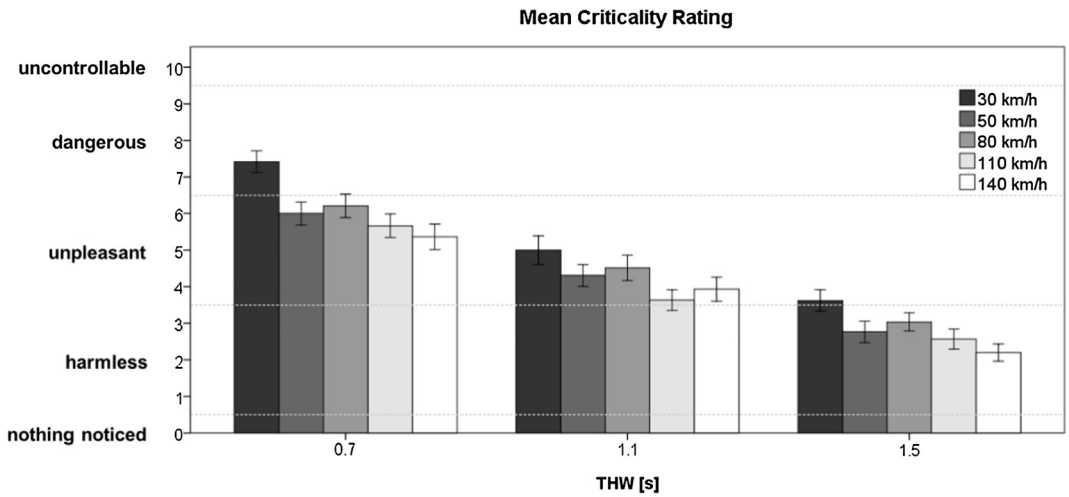

Fig. 3. Mean ratings and standard errors for all three time headways and five velocities. The simulated driving environment for 30 km/h and 50 km/h was an urban road, for 80 km/h and 110 km/h a rural road and for 140 km/h a national highway.

Table 2
Subjective risk ratings per scenario, mean and [standard deviation].

	Velocity				
	30 kmh	50 kmh	80 kmh	110 kmh	140 kmh
Time headway					
0.7 s	7.41 [1.59]	6.00 [1.69]	6.21 [1.72]	5.67 [1.75]	5.37 [1.92]
1.1 s	5.00 [2.10]	4.31 [1.54]	4.52 [1.86]	3.63 [1.54]	3.93 [1.80]
1.5 s	3.62 [1.59]	2.77 [1.59]	3.03 [1.35]	2.57 [1.50]	2.20 [1.30]

To statistically test the hypothesis that not only THW but also a velocity influence on subjective risk, a repeated measures ANOVA with the main factors "THW" and "velocity" and the two-way interaction between "THW" and "velocity" was conducted. Since both factors violated the assumption of sphericity, degrees of freedom were adjusted following the Greenhouse-Geisser-procedure. As expected, mean ratings differ significantly as a function of "THW" ($F(1.70, 47.67) = 188.23$, $p < .001$, $\eta_p^2 = 0.87$) and "velocity" ($F(2.80, 78.48) = 12.97$, $p < .001$, $\eta_p^2 = 0.32$). No significant interaction between both factors can be found ($F(5.6, 155.7) = 1.09$, $p = .37$, $\eta_p^2 = 0.04$). Two planned contrasts were to test hypotheses that the velocity effect should be found even though the driving environment was held constant. Therefore, in urban and rural environments, the lower velocity was tested against the higher velocity (urban: 30 km/h vs. 50 km/h; rural: 80 km/h vs. 110 km/h). Planned contrasts revealed that drivers rate the same driving environments and THWs as more critical when following with a lower velocity compared to a higher velocity. This applies for urban areas ($F(1, 28) = 19.04$, $p < .001$, $\eta^2 = 0.40$), and for rural roads ($F(1, 28) = 18.77$, $p < .001$, $\eta^2 = 0.51$).

One of the major advantages of the scale used in this experiment is the clear definition of a threshold for critical situations (0–6: no critical situation; 7–10: critical situation). A binary logistic regression was conducted with "THW", "velocity", and the interaction term as predictors and a binary term (below or over the criticality threshold) as the dependent variable. Testing the full model against a constant only model reveals a significant difference, $X^2 = 118.57$, $p < .001$. Nagelkerke's R^2 of 0.39 indicates a weak to moderate relationship between predictors and grouping. Using the Wald criterion, only the predictor "THW" reaches a level of significance, $p = .001$, $Exp(B) = 0.012$. "Velocity" and the interaction term are no significant predictors, $p = .62$ and $p = .60$. Fig. 4 shows the predicted probabilities for critical ratings (SBS > 6). Note that the effect of velocity on subjective risk (yes/no) descriptively is dependent on the driven THW, with a steep curve for smaller THWs and a flat curve for larger THWs.

4. Discussion

In this motion-based driving simulator study, twenty-nine participants rated all combinations of five velocities and three THWs regarding their subjective risk and perceived criticality. Based on on-road data (Brackstone et al., 2009), physical calculations (Winner, 2009), and psychological driving theories (Fuller, 2005; Wilde, 1982) it was hypothesized that the subjective risk of each car following scenario would be influenced not only by THW but also by velocity. Compared to previous studies, a more realistic and natural driving task and environment was used due to an active role of the driver in the driving task and a velocity-adapted driving environment.

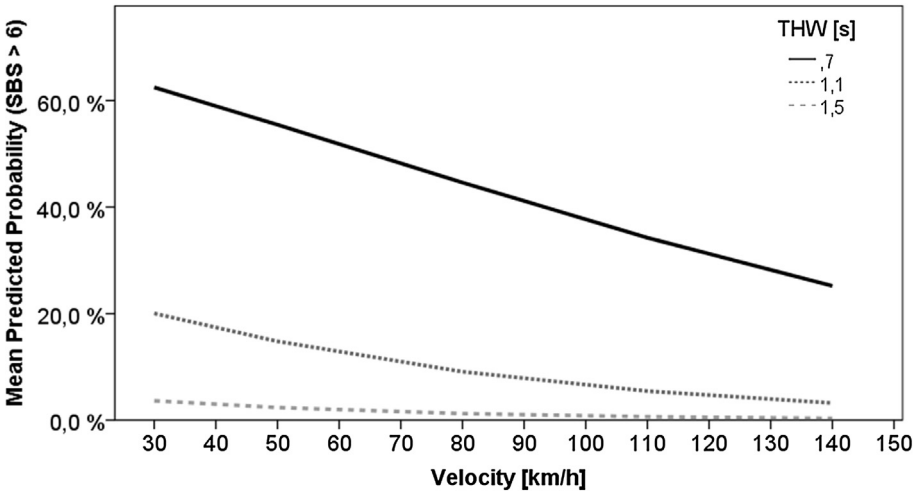

Fig. 4. Results of the binary logistic regression model. Predicted probabilities for ratings over 6 points of different velocities and THWs.

The here presented work provides evidence for a velocity effect on the subjective risk during car following scenarios previously reported only in real-traffic studies. Consistent with literature (Siebert et al., 2014, 2017), smaller THWs were rated as significantly more critical on a standardized subjective risk scale. Additionally, results indicate that over all conditions, the same THWs were rated as significantly more critical by the drivers at lower velocities compared to higher velocities. To eliminate one possible confounding variable that could have led to the absence of the velocity effect in previous studies (e.g., Siebert et al., 2014, 2017), car following scenarios were tested in driving environments realistic for the given velocity (urban, rural, and highway). A detailed analysis of these driving environments (urban and rural), each comparing a higher and a lower velocity, also revealed a velocity effect. This is interesting since it suggests that the velocity effect might not be based only on different driving environments of different travel velocities alone. Still, the predicted probabilities of the logistic regression show that within the very narrow frame of investigated THWs between 0.7 s and 1.5 s, the influence of velocity on subjective risk is descriptively linear but not significant. A threshold effect as suggested for the influence of different THWs was not found for the velocity effect.

These results could partially explain on-road findings, where drivers prefer larger THWs at lower velocity ranges and lower THWs at higher velocity ranges (Brackstone et al., 2009; Knospe et al., 2002). Since the same THWs are perceived as more critical at lower velocities, drivers might adapt their behavior and prefer larger THWs at lower velocities.

On the other hand, it contradicts findings from simulator studies that did not report any influence of velocity on the preferred THWs or subjective risk (Lewis-Evans & Rothengatter, 2009; Siebert et al., 2014; Taieb-Maimon & Shinar, 2001; van Winsum & Heino, 1996). Several issues could be held responsible for this. First, in the present study, participants followed the preceding vehicle freely and except of the instruction for experimental manipulation of the THW, their actual behavior was not constrained by automated systems that take over longitudinal control of the host vehicle (e.g., ACC). This outlines a crucial difference of the present study to other studies in which participants rated scenarios generated by highly automated driving (Lewis-Evans & Rothengatter, 2009; Siebert et al., 2014) or video simulations (Fuller et al., 2008). Second, in this study, the driving environment was not constant as in Siebert et al. (2014) but adapted to the velocity range (urban, rural, highway). As discussed in the beginning, different driving environments could influence the perceived criticality during car following and the velocity effect itself could play a minor role. However, the perceived criticality significantly differed between velocities also within the same environments. Contrary to previous studies, only driving environments that are more realistic to the single velocities might have revealed the velocity effect. This is in line with the assumption of Siebert et al. (2014) who also found lower criticality ratings for the speed that drivers usually choose in the driving environment (rural road, 100 km/h). Generic vs. realistic driving environments might be the core difference to previous studies, which might have covered the velocity effect until now. Third, slightly different subjective ratings were applied that need to be considered when comparing the results of this study with findings reported in the literature. Besides the criticality scale that was also used in this study, Siebert et al. (2014) included more measures on risk perception (e.g., criticality scale, Likert scales on risk, task difficulty). However, they state that, in their study, "all reported subjective variables on the different time headways and speeds were significantly correlated" (p. 69). The inclusion of more subjective measurements might, therefore, contribute only little additional insights.

Besides these differences, the results of this study are partly consistent with these previous findings. The clear relation between THW and criticality ratings could be replicated with the ERS-based method rendering it a valid approach for investigating car following behavior. From a methodological perspective, this study thus also contributes a new method for the experimental evaluation of different following distances during car following. Another difference to previous studies is

the age distribution of the sample. It has been reported that age is an essential parameter for preferred THWs (Evans & Wasielewski, 1983; Ohta, 1993) and that some effects (e.g., daytime) only affect preferred THWs of older but not younger drivers (Kiefer et al., 1999). Still, most studies reported in the literature included only young drivers with average ages between 25 and 30 years (e.g., Siebert et al., 2014; van Winsum & Heino, 1996). The present study used a sample ranging from 22 to 61 years, thus covering a larger age-range. The influence of age on the velocity effect remains an interesting question for future studies. Also, some of the referenced studies did only investigate very narrow velocity ranges, for example, 40–70 km/h (van Winsum & Heino, 1996). Investigating a broader velocity range as in the present study renders the finding of an actual velocity effect more likely. One could also argue, that the color of the ERS (yellow, grey, blue) influenced the criticality ratings of the participants (see also Löffler, Paier, Toriizuka, Ikeda, & Hurtienne, 2015). Still, the colors were held constant over all velocities, and our results are consistent with other studies regarding the effect of THW (Siebert et al., 2014). One crucial issue must be considered when using the ERS method with the same THWs in different velocities: task demand. Even though the tolerance range was constant in terms of THW-seconds, different velocities implied different tolerance ranges in meters. Additionally, the ERS method draws the driver's visual attention to the space between the two vehicles and increase the necessary monitoring frequency to detect relevant events in the driver's environment (which would be more relevant in dense urban environments than in rural areas or the highway). This could have led to different task demands during car following. For example, keeping the instructed THW could have required more frequent and more intense braking in lower velocity ranges (with smaller tolerance in meters) compared to higher velocity ranges (with larger tolerances in meters). However, in a pre-study with 10 participants (instruction, procedure, and manipulation of the three THWs and the five velocities almost identical to this study), the usage of brake pedal was analysed for a pre-defined point on the road during car following with the ERS. Here, participants did not differ in their brake pedal usage between velocity ranges for the same THWs ($F(4, 390) = 1.055$, $p = .379$) indicating that different velocities did not require more or more intense speed adjustments. A major bias on account of the ERS method is therefore not to be excluded, but highly unlikely.

Contrary to the results' general trend of decreasing subjective risk ratings for increasing velocities, all three THWs show a slight increase of subjective risk ratings when velocity increases from 50 km/h to 80 km/h. Still, subjective risk ratings decrease again when velocity increases from 80 km/h to 110 km/h. One explanation for this exception from the general pattern could be the different driving environments. The 50 km/h scenarios took place in an urban area, a normal velocity for German drivers in this type of road. For rural roads, 80 km/h is a rather low velocity and might have been considered as more "unsafe" compared to the here normal traveling speed of 110 km/h. A similar effect was also reported by Siebert et al. (2014), where deviations from a "standard" velocity are presumed to increase subjective risk ratings of the scenario possibly.

In summary, this study adds to existing literature on car following behavior by providing evidence for the velocity effect on subjective risk ratings in an experimental setup with velocity-realistic driving environments. The results suggest that different preferred THWs for different velocities – a phenomenon observed in on-road data – could be explained via an increased subjective risk for lower velocities. Future studies should investigate systematically how this effect is affected by the level of automation and driver parameters as, for example, age.

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